

Biotechnology Companies among the KOSDAQ-Listed Special Technology Companies

The “special listing track for technology growth companies” has been launched to allow firms with strong technological capabilities and growth potential to access the capital market, even if their current operating performance is limited.

Of the 38 companies newly listed through the technology-special listing program in 2025, 17 (44.7%) were bio/healthcare-related companies, up from 18 companies (40%) in 2024.

The biotechnology companies newly listed through the technology-special listing program in 2025 were primarily engaged in next-generation modality-based drug development and platform technologies*, biomaterials**, medical devices, medical supplies, and analytical services***.

* Including immune cell therapies for cancer, ADC-based anticancer therapeutics and platform technologies, and regenerative therapeutics

** Including organoid-, microbiome, and exosome-based biomaterials

*** Including medical AI technologies, medical supplies and diagnostics, and biomarker-based analytical services

🔍 Newly Listed Technology Special Companies by Technology Area, (2005-2025)

(Unit: No. of companies)

Classification	2005 - 2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Biotechnology	24	9	5	18	17	15	14	10	14	18	17
Non-Biotechnology	3	2	3	4	7	11	25	23	22	27	21
Total	27	11	8	22	24	26	39	33	36	45	38

Note 1) Includes companies listed under the Korea Exchange (KRX) KOSDAQ Technology Growth Division with a history of technology-special listing. Their current listing status (e.g., transfer to another market segment such as the Premier Division, or designation as an Issue under Management under KRX regulations) is not reflected.

Note 2) Classification of bio/healthcare companies was conducted independently by the National Biotechnology Policy Research Center (NBPRC) based on each company's major products and services. The classification covers companies across the entire biotechnology sector, including red, green, and white biotechnology.

Source: KOSDAQ Technology Growth Division, Korea Exchange (KRX) as of February 20, 2026; BiolNwatch Vol. 26-17, National Biotechnology Policy Research Center (NBPRC), March 12, 2026

Strategy for Fostering AI-Driven Biotechnology

Throughout 2025, the Korean government progressively advanced its strategy to develop the bio sector as the “next strategic industry after semiconductors.”

The “AI Bio Diffusion Strategy” presented the technological approach, while the “National AI Bio Strategy” finalized the overall framework by defining key implementation actors and priority focus areas.

While initially aiming for a “quantum leap” across the entire bioindustry, the strategies significantly enhanced policy effectiveness by establishing:

- a clear execution body, the “National AI Bio Institute,”
- an advanced technological paradigm centered on agentic AI, and
- clearly defined targets through five key priority sectors.

Strategy (Date of Announcement)	AI Bio Diffusion Strategy (Apr. 30, 2025)	National AI Bio Strategy (Dec. 18, 2025)
Lead Agency	Ministry of Science and ICT (MSIT) (Council for Comprehensive Biotechnology Policy)	Jointly led by MSIT and other relevant ministries (Ministerial Meeting on Science and Technology)
Key Features	Technology Diffusion & Foundational Building Transformation of the research paradigm through the adoption of AI technologies Focus on securing and disseminating core technologies to shift bio research toward an AI-centered paradigm	Implementation & Advancement Establishment of key hubs and focused development of five priority areas Selection of five key priority sectors and establishment of dedicated research institutes to generate tangible outcomes through ‘AI-Bio’ convergence and elevate the approach to a national strategy
Vision	Future Bio Innovation Driven by AI	Advancing as a Global Hub for AI-Driven Biotechnology
Key Objectives	Securing core AI technologies Building autonomous laboratories Fostering interdisciplinary talent	Establishing the National AI Bio Institute Driving innovation in five key priority sectors Implementing a “Lab-in-the-Loop” system

Source: Ministry of Science and ICT (MSIT)

Fostering the Bioindustry as a New Growth Engine through Comprehensive Support

Establishing a Comprehensive Support System at the National Level



Launch of the National Bio Innovation Committee

Establishment of an integrated bio policy central coordinating body under the Prime Minister; Announcement of the "Bioindustry Policy Roadmap" in Q1 2026

Record-Level Investment

Deployment of KRW 2.3 trillion through the National Growth Fund; Creation of a KRW 60 billion Phase III clinical trial-focused fund to support drug development targeting the global market

Regulatory Innovation

Significant reduction in approval timelines for new drugs and medical devices (from 420 days to 240 days); Establishment of Phase III exemption criteria for biosimilars

Qualitative Leap through AI-Bio Convergence



AI Bio Innovation Hub

Application of regulatory sandboxes for data utilization; Establishment of an integrated National Bio Data System to enable data sharing and utilization

Adoption of AI in Healthcare

Development of a healthcare data space (by the 1st half of 2026); Phased expansion of AI-based drug review services (from generics to new drugs)

R&D in Six Promising Areas

End-to-end support from development to clinical trials and approval for advanced medical devices, including medical robots and implants

* Six promising areas: Software as a Medical Device (SaMD), u-Healthcare, medical robots, implants, interventional medical devices, and next-generation molecular diagnostics

Accelerating Global Expansion



Development of new export models based on overseas acquisitions and hospital expansion abroad

Advancement of clusters such as Advanced Medical Complexes

Expansion of the use of health and medical data and digital technologies through the enactment of the Digital Healthcare Act (bill proposed on Nov. 24, 2025)

Laying the Foundation for a Bioeconomy

Enactment of the “Synthetic Biology Promotion Act”

The Synthetic Biology Promotion Act, enacted in April 2025, establishes a national-level strategic framework to foster the field of synthetic biology, which creates new biological systems through DNA design and synthesis.

Amid intensifying global competition for technological leadership, the Act institutionalizes synthetic biology as a national strategic technology and provides a legal foundation for the transition toward a data- and AI-driven digital bio-manufacturing paradigm.

The Act serves as a critical institutional cornerstone for Korea’s advancement as a global leader in synthetic biology and represents a key piece of legislation ushering in the era of AI–bio convergence.

Establishment of a National Biofoundry

Provides the legal basis for building infrastructure to accelerate bio research and drive innovation in industrial bioprocessing

Biosecurity and Biosafety Management

Promotes a responsible innovation ecosystem alongside technological advancement



Introduction of a “Fast-Track Confirmation System”

Reduces regulatory uncertainty and enables the rapid commercialization of research outcomes

Integrated Support for Infrastructure Talent, and Technology Development

Cultivates a skilled workforce in synthetic biology and accelerates innovation in bio-manufacturing

Enactment of the “Special Act on Regulatory Support for Bio-Pharmaceutical Contract Development and Manufacturing Organizations (CDMOs)”

The CDMO Special Act, passed by the National Assembly in December 2025, streamlines the regulatory framework for domestic CDMO companies by introducing a registration system for export-oriented manufacturers, establishing a legal basis for GMP compliance certification, and implementing a certification system for raw materials.

By consolidating regulations that were previously dispersed across multiple laws—such as the Pharmaceutical Affairs Act and the Advanced Regenerative Bio Act—into a single legislative framework, the Act lays out, for the first time, an institutional foundation to enhance the global credibility and international competitiveness of Korean CDMOs.

Establishing a Foundation for the Integration and Utilization of Data in the Bio Sector (Planned)

(Tentative)

Bio Data Act – Bill Proposed

AI-Bio convergence is a key driver of innovation in drug development and precision medicine, and access to high-quality bio data is critical to its success.

At present, data remains fragmented across ministries and institutions, making integrated utilization difficult. Even outcomes from publicly funded national R&D projects are not systematically shared. In addition, rigid regulatory frameworks—such as the Bioethics and Safety Act and the Personal Information Protection Act—as well as complex IRB procedures, constrain the utilization of bio data for research purposes.

The proposed Act seeks to establish a standardized and integrated framework for bio data management and introduce reasonable special regulatory provisions, thereby fostering an AI-Bio innovation ecosystem that can shorten drug development timelines and advance precision medicine.

Ultimately, the Act aims to generate tangible public benefits and contribute to both national welfare and economic growth.

Source: Proposed Bill on the Promotion of Bio Data Utilization and AI-Bio R&D, Legislative Status of the National Assembly, National Legislation Information Center, Ministry of Government Legislation, Dec. 24, 2025

(Tentative)

Digital Healthcare Act – Bill Proposed

Korea became a super-aged society in December 2024, with more than 20% of its population aged 65 or older. As a result, the prevalence of chronic diseases and healthcare expenditures for senior adults are rising sharply. At the same time, advances in ICT—including IoT and AI—have made real-time monitoring of chronic conditions such as diabetes, cardiovascular disease, and stroke feasible.

The proposed Act on the Promotion of Digital Healthcare and the Utilization of Health and Medical Information aims to improve the quality of healthcare services and promote public health through the use of health and medical information and digital technologies, while also enabling their broader application across patient care, research, and industry.

Source: Proposed Bill on the Promotion of Digital Healthcare and the Utilization of Health and Medical Information, Legislative Status of the National Assembly, National Legislation Information Center, Ministry of Government Legislation, Nov. 24, 2025